Northgate Steel Mill (fictional)

INSPECTION CERTIFICATE EN 10204 - 3.1

Mill test certificate · sections coded per EN 10168

Certificate no.	**NSM-26-48817**
Date of issue	**2026-08-21**
Page	**1 / 1**

A COMMERCIAL TRANSACTION

A01 Manufacturer	A02 Type of document
Northgate Steel Mill (fictional)	**EN 10204:2004 type 3.1**
A06 Purchaser	**A07 Purchase order**
Kaneda Fabrication Works (fictional)	**PO-26-0771 · Job 26-118**

B DESCRIPTION OF PRODUCT

B01 Product	B02 Steel grade / specification	
Plate, normalized	**ASME SA-516/SA-516M Gr.70 (2023), normalized**	
B07 Heat (cast) no.	**B09-B11 Dimensions**	**B13 Quantity**
4A7721	**20 x 2500 x 8000 mm**	**3 plates, 9.42 t**
B03 Delivery condition		**Nominal thickness (mm)**
Normalized 900 °C, 30 min, air cooled		**20.0**

C INSPECTION - MECHANICAL TESTS (C00-C69)

Test	C11 Yield ReH (MPa)	C12 Tensile Rm (MPa)	C13 Elongation A (%)	Direction / location
Required	260 min.	485 - 620	21 min.	
Result, heat 4A7721	345	520	26	Transverse, 1/4 t

C40 Charpy V (J)	C42 Test temp (°C)	C43 Single values (J)	Average (J)	Required (J)
Transverse, 10x10	-29	64 / 58 / 66	62	20 (average, CVN 10x10)

C CHEMICAL COMPOSITION - LADLE ANALYSIS, % BY MASS (C70-C92)

Heat	C	Si	Mn	P	S	Cr	Ni	Mo	Cu	V	Nb
4A7721	.18	.35	1.10	.012	.004	.02	.02	.01	.03	.002	.001

Carbon equivalent CE(IIW) = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15 = 0.37

D OTHER TESTS

D01 Marking, visual inspection and dimensional check: satisfactory. Heat number hard-stamped on each piece.

D02 Ultrasonic examination per SA-435: satisfactory.

Intended use (per purchase order): Shell and heads (2:1 SE), vessel V-110.

Z VALIDATION

Z01 We hereby certify that the products described above comply with the terms of the order and the material specification, based on specific inspection of the products supplied.

Z02 Date of issue	Z03 Authorized inspection representative (independent of manufacturing)
2026-08-21	**S. Varga (fictional), QA** *signed S. Varga* Stamp QA-07

SYNTHETIC DOCUMENT - generated for software testing - not a real record